

Single Phase (1 ϕ) AC Electrical Energy

AC (alternating current) electrical energy, both single phase and three phase, is the most widely used form of electrical energy in electric power systems and electric machinery applications. It is a more complicated form of electrical energy than DC because by its nature, all voltages and currents associated with AC electrical energy are sinusoidally varying with time. The sinusoidal frequency (usually 50HZ or 60HZ) of AC electrical energy is constant. The mathematical expressions below define voltage and current at an AC electrical energy interface.

$$\begin{aligned}v(t) &= V_p \cos(\omega t + \alpha) & i(t) &= I_p \cos(\omega t + \beta) \\ \omega &= 2\pi f = \text{oscillating frequency in radians per second} \\ f &= \text{oscillating frequency in Hertz (cycles per second)}\end{aligned}$$

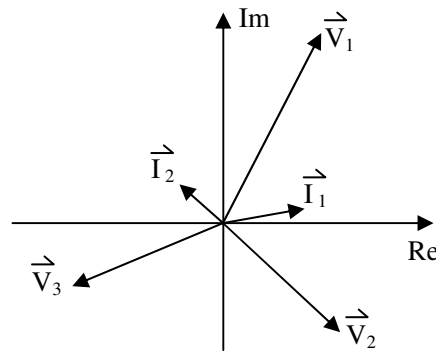
AC electrical energy is assumed to be steady state, which means that the frequency of oscillation is constant, and also that the magnitudes and relative phase angles of the voltage and current are constant and do not change with time. So although AC currents and voltages are functions of time, they are functions of time in a very constant and predictable way. This allows them to be more conveniently mathematically represented in a special way called *phasors*, in which the time and frequency variables are not considered. If the time and frequency variables are not considered then that leaves only the two variables, magnitude and phase angle. Therefore *phasors* are a special mathematical way of representing AC voltages and currents using only their magnitudes and phase angles. This makes phasors mathematically identical to 2-D vectors or complex numbers, and that is exactly how they are worked with in AC electrical calculations. The correlation between time domain AC voltage/current and their mathematical representation as phasors is defined below.

$$\begin{aligned}v(t) &= V_p \cos(\omega t + \alpha) \rightarrow \bar{V} = V \angle \alpha \quad \text{where} \quad V = V_p / \sqrt{2} \\ i(t) &= I_p \cos(\omega t + \beta) \rightarrow \bar{I} = I \angle \beta \quad \text{where} \quad I = I_p / \sqrt{2}\end{aligned}$$

Notice above that vector notation (arrowhead symbol) is used with phasors. This is appropriate since a phasor is a 2-D quantity that has both a magnitude and phase angle. Also the vector notation is appropriate because a phasor is also a complex variable having both a magnitude and a phase angle (or a real part and imaginary part). Most textbooks use bold face notation when writing phasor variables. Since it is difficult for humans to hand write in bold face, we instead use the vector (arrowhead symbol) notation for phasors. It should be understood that this **vector notation must be used** when working with phasors, and **it is not optional**. This is very important because the same variable names are often used for the phasor and its magnitude, and the vector notation is the only way of telling them apart.

Also notice above that peak value variables V_p and I_p are used in the time domain expressions, but unsubscripted variables V and I are used for the phasor magnitudes. These unsubscripted variables V and I are rms (root mean square) values, and they are related to the peak values by a simple square root of two factor as shown. It is commonly assumed that **AC electric power and machinery applications always use rms values in phasors and in power calculations and never peak values**. The only time peak values are used is when writing time domain expressions for voltage and current, and this is something that is rarely required in AC power calculations.

When working with phasors, it is useful to visualize them using a *phasor diagram*. A *phasor diagram* is just like simple vector diagram in that it is a special type of 2-D plot. A *phasor diagram* is created by first drawing horizontal and vertical axes, and then labeling the horizontal axis with Re (for real part axis) and the vertical axis with Im (for imaginary part axis). Phasors are then drawn in as vectors starting at the origin and having a specific length determined by the phasor's magnitude and in a specific direction determined by the phasor's phase angle. The real axis is the zero degree reference for phase angles. Positive phase angles are defined in a counterclockwise direction from the real axis, and negative phase angles are defined in a clockwise direction. This is illustrated in the example below.



$$\begin{aligned}\vec{V}_1 &= 120 \angle 60^\circ \text{ V} \\ \vec{V}_2 &= 100 \angle -45^\circ \text{ V} \\ \vec{V}_3 &= 90 \angle -150^\circ \text{ V} \\ \vec{I}_1 &= 23 \angle 15^\circ \text{ A} \\ \vec{I}_2 &= 20 \angle -135^\circ \text{ A}\end{aligned}$$

Of great importance in AC power calculations is the phase angle difference between two phasors, particularly between a phasor voltage and a phasor current at an AC interface. **The phase angle difference between two phasors has both a magnitude and an orientation.** It is determined by first choosing one of the phasors as a reference and then using a magnitude and orientation to describe where the other phasor is located with respect to the first one. The orientation is a one word verbal descriptor (leading, lagging, or unity) which specifies where the second phasor is in time with respect to the first phasor.

For example looking at phasors $\vec{V}_1$ and $\vec{I}_1$, the phase angle difference magnitude between them is obviously 45° , but what is the orientation? Remember that the orientation is determined by choosing one phasor as a reference and then describing where the other phasor is in time with respect to the first one. The best way to visualize time passing on a phasor diagram is to imagine the second hand of a clock being superimposed on the stationary phasor diagram and moving in a clockwise direction past the phasors. As time passes, this imaginary second hand passes by phasors in a certain order, and this is how the orientation in time between any two phasors is described. Again looking at phasors $\vec{V}_1$ and $\vec{I}_1$, one can say that $\vec{V}_1$ happens earlier in time than $\vec{I}_1$, and is said to be leading $\vec{I}_1$ by 45° . One can also say that $\vec{I}_1$ happens later in time than $\vec{V}_1$, and is said to be lagging $\vec{V}_1$ by 45° . Both statements are correct and are just a matter of what phasor is the chosen reference.

A **very important rule** that should be followed when making a leading/lagging determination between two phasors is that **the angle of comparison must be less than 180°** . For example looking at phasors $\vec{V}_1$ and $\vec{V}_3$, one might try to say that $\vec{V}_3$ lags $\vec{V}_1$ by 210° , but this would not be an appropriate statement since the angle of comparison is greater than 180° . The correct way to describe the phase angle difference between those two phasors would be to say that $\vec{V}_1$ lags $\vec{V}_3$ by 150° or $\vec{V}_3$ leads $\vec{V}_1$ by 150° . For practice, try picking any other pair of phasors in the previous example and making a leading/lagging determination.

By definition there are three basic operating variables at a single phase AC electrical energy interface or port: *AC voltage rms magnitude*, *AC current rms magnitude*, and *power factor angle*. This is illustrated in the single phase AC ports of the two diagrams below, where the port parameters are shown.



The port variable defined using the Greek letter θ (theta) is the *power factor angle*. By definition, the power factor angle is angle by which the AC current lags the AC voltage in time. However, the easiest way to remember the definition of the *power factor angle* is by its mathematical relationship to the voltage and current phasors. The *power factor angle* is always the voltage angle minus the current angle.

$$\begin{aligned}\text{If } \vec{V} &= V \angle \alpha \text{ and } \vec{I} = I \angle \beta \text{ then} \\ \theta &= \text{pf angle} = (\text{voltage angle}) - (\text{current angle}) = \alpha - \beta\end{aligned}$$

In AC power calculations, **power is not just voltage times current like it is for DC power**. With AC there are more parameters to account for, and consequently there is more than one type of power which can be defined and calculated. There are actually five different types of power defined and worked with. They are instantaneous power, real power, reactive power, complex power and apparent power. However the AC power quantities of most interest are usually real power and reactive power. The table below defines the equations and units of the five different 1 ϕ AC power quantities.

Quantity	Symbol	Equation(s)	Units
Instantaneous Power	$p(t)$	$p(t) = v(t) \cdot i(t) = V_m \cos(\omega t + \alpha) \cdot I_m \cos(\omega t + \beta)$ $= (VI \cos \theta)(1 + \cos 2\omega t) + (VI \sin \theta)(\sin 2\omega t)$	Watts (W)
Real Power	P	$P = VI \cos \theta$	Watts (W)
Reactive Power	Q	$Q = VI \sin \theta$	Volt-Amperes Reactive (VAR)
Complex Power	$\bar{S}$	$\bar{S} = P + jQ = S \angle \theta$ $\bar{S} = \bar{V} \cdot \bar{I}^*$ in an equivalent circuit $P = \text{Re}\{\bar{S}\}$ and $Q = \text{Im}\{\bar{S}\}$	Volt-Amperes (VA)
Apparent Power	S	$S = \bar{S} = VI = \sqrt{P^2 + Q^2}$	Volt-Amperes (VA)
Power Factor	pf	$pf = \cos \theta = \cos[\tan^{-1}(Q/P)]$	Unity, Leading, or Lagging $(\theta=0), (\theta<0), (\theta>0)$

A special calculation associated with AC power is the *power factor*. The *power factor* is by name obviously related to the power factor angle, and it can be thought of as just a different way of expressing the power factor angle. However, **a power factor is expressed using two distinct parts**, magnitude and orientation, and **it is incomplete and incorrect without both parts**. The magnitude is the cosine of the power factor angle θ , and the orientation is a verbal descriptor (leading, lagging, or unity) which is determined by the sign of the power factor angle θ . A power factor is leading if θ is negative, lagging if θ is positive, and unity if θ is zero.

Example #1:

A single phase AC generator operates with an output of 120V and supplies 20KVA at a 0.5 leading power factor to a system. Calculate the generator output current and the real & reactive power output power.

$$V = 120 \text{ V}, \quad S = 20 \text{ KVA}, \quad pf = 0.5 \text{ leading} \rightarrow \theta = -\cos^{-1} 0.5 = -60^\circ$$

$$I = \frac{S}{V} = \frac{20,000}{120} = 166.66 \text{ A} \leftarrow \text{source current}$$

$$P = S \cos \theta = (20,000) \cos(-60^\circ) = 10,000 \text{ W} = 10 \text{ KW} \leftarrow \text{generator real power output}$$

$$Q = S \sin \theta = (20,000) \sin(-60^\circ) = -17,320.51 \text{ VAR} = -17.3205 \text{ KVAR} \leftarrow \text{generator reactive power output}$$

Example #2:

A single phase AC source operates with an output of 400V and supplies 30KW at a 0.9 lagging power factor to a system. Calculate the source output current and the complex power output.

$$V = 400 \text{ V}, \quad P = 30 \text{ KW}, \quad pf = 0.9 \text{ lagging} \rightarrow \theta = +\cos^{-1} 0.9 = 25.842^\circ$$

$$P = VI \cos \theta \rightarrow I = \frac{P}{V \cos \theta} = 83.33 \text{ A} \leftarrow \text{source output current}$$

$$S = VI = 33,333.33 \text{ VA} = 33.33 \text{ KVA}$$

$$\bar{S} = S \angle \theta = 33.33 \angle 25.842^\circ \text{ KVA} \leftarrow \text{complex power output of the source}$$

Example #3:

A single phase AC load consumes 10KW and 2KVAR while operating at 160V. Calculate the load input current, power factor, and the equivalent impedance of the load.

$$V = 160 \text{ V}, \quad P = 10 \text{ KW}, \quad Q = 2 \text{ KVAR}$$

$$\bar{S} = P + jQ = (10,000) + j(2,000) = 10,198.03903 \angle 11.310^\circ \text{ VA}$$

$$S = 10,198.03903 \text{ VA}, \quad \theta = 11.310^\circ$$

$$I = \frac{S}{V} = \frac{10,198.03903}{160} = 63.738 \text{ A} \leftarrow \text{load input current}$$

$$pf = \cos \theta = 0.980581 \text{ lagging } (\theta > 0) \leftarrow \text{load input power factor}$$

$$\bar{Z} = Z \angle \theta = \left(\frac{V}{I} \right) \angle \theta = \left(\frac{160}{63.738} \right) \angle 11.310^\circ = 2.5103 \angle 11.310^\circ \Omega \leftarrow \text{equivalent impedance of the load}$$

Example #4:

A single phase load input terminals operates at 12KV, 78A, and a 0.754 lagging power factor. Calculate the real and reactive power input to the load and the equivalent impedance of the load.

$$V = 12 \text{ KV}, \quad I = 78 \text{ A}, \quad pf = 0.754 \text{ lagging} \rightarrow \theta = +\cos^{-1} 0.754 = 41.062^\circ$$

$$P = VI \cos \theta = (12,000)(78) \cos 41.062^\circ = 705.7440 \text{ KW} \leftarrow \text{load input real power}$$

$$Q = VI \sin \theta = (12,000)(78) \sin 41.062^\circ = 614.8345 \text{ KVAR} \leftarrow \text{load input reactive power}$$

$$\bar{Z} = Z \angle \theta = \left(\frac{V}{I} \right) \angle \theta = \left(\frac{12,000}{78} \right) \angle 41.062^\circ = 153.8462 \angle 41.062^\circ \Omega \leftarrow \text{equivalent impedance of the load}$$

Just as Ohm's Law yields additional power equations (I^2R and V^2/R) in DC, the same is true with AC. Ohm's Law in an AC circuit states that the ratio of the phasor voltage at a port to the phasor current passing through the port is defined as the equivalent impedance of the circuit as seen at the port. This can be more simply stated mathematically by the following complex variable equations.

$$\bar{Z} = \bar{V} / \bar{I} \quad \text{or} \quad \bar{V} = \bar{Z} \bar{I} \quad \text{or} \quad \bar{I} = \bar{V} / \bar{Z}$$

$$\text{If } \bar{V} = V \angle \alpha \quad \text{and} \quad \bar{I} = I \angle \beta \quad \text{then}$$

$$\bar{Z} = \bar{V} / \bar{I} = (V \angle \alpha) / (I \angle \beta) = (V/I) \angle (\alpha - \beta) = (V/I) \angle \theta = Z \angle \theta$$

Notice in the above equations that the magnitude of the impedance is related to the magnitudes of the voltage and current phasors. Also notice that **the impedance angle is equal to the power factor angle**. This should always be remembered when doing AC power calculations at impedance loads since it can often simplify the calculations.

Combining Ohm's Law with the basic definitions of AC power quantities yields several additional forms of the basic AC power equations which are shown below. These extra equations are very useful when doing load power calculations.

$$P = VI \cos \theta = (V^2/Z) \cos \theta = I^2 Z \cos \theta, \quad Q = VI \sin \theta = (V^2/Z) \sin \theta = I^2 Z \sin \theta$$

$$S = VI = V^2/Z = I^2 Z, \quad \bar{S} = \bar{V} \cdot \bar{I}^* = V^2/\bar{Z}^* = I^2 \bar{Z}$$

With AC sources, one does not normally work with an impedance variable since it does not have physical significance or meaning at a source. However the impedance variable does exist for an AC load, and it should always be interpreted as the equivalent impedance of the load as seen by what is supplying energy to it.

Three Phase (3 ϕ) AC Electrical Energy

The electric utility company power grids used throughout the world are for the most part three phase AC. Parts of all electric utility company power grids are also single phase AC, but only at the residential and small commercial level. Industrial and commercial electrical energy applications all use three phase AC power.

Fortunately, once single phase AC electrical energy is understood then three phase electrical energy can be more easily explained and accepted. The primary difference between the definitions of single phase and three phase systems lies in the definition of a port in those systems. In single phase systems, a port has two terminals, one phasor voltage, and one phasor current associated with it. In three phase systems, a port has three or four terminals (A,B,C or A,B,C,N), three or four phasor currents, and three or six phasor voltages associated with it. This is illustrated in the three phase AC ports of the two diagrams below.



At a three phase AC port, the terminals A,B,C are called the *line terminals*, and the terminal N is called the *neutral terminal*. The four phasor currents at a three phase port flow through these four terminals and are called the *line currents* and the *neutral current*. The six phasor voltages at a three phase port are broken up into two logical groups of three quantities and are called the *line to line voltages* and the *line to neutral voltages*. Therefore at a three phase port, there are a total of ten phasor voltage/current quantities that are the basic operating variables of the port. These ten phasors at a three phase port and how they are commonly written in sets of quantities is shown below.

L-N Voltages	L-L Voltages	Line Currents	Neutral Current
$\begin{cases} \vec{V}_{AN} = V_{AN} \angle \alpha_{AN} \\ \vec{V}_{BN} = V_{BN} \angle \alpha_{BN} \\ \vec{V}_{CN} = V_{CN} \angle \alpha_{CN} \end{cases}$	$\begin{cases} \vec{V}_{AB} = V_{AB} \angle \alpha_{AB} \\ \vec{V}_{BC} = V_{BC} \angle \alpha_{BC} \\ \vec{V}_{CA} = V_{CA} \angle \alpha_{CA} \end{cases}$	$\begin{cases} \vec{I}_A = I_A \angle \beta_A \\ \vec{I}_B = I_B \angle \beta_B \\ \vec{I}_C = I_C \angle \beta_C \end{cases}$	$\vec{I}_N = I_N \angle \beta_N$

Real world three phase power systems and electric machines that used three phase electrical energy are designed and operated in what is called a *balanced and symmetric mode*. Balance is a constraint that makes the magnitudes of all three phasors within each set of quantities equal. Symmetry is a constraint that makes the phase angles of all three phasors within each set of quantities equally spaced from one another in time by 120° and in the time sequence ABC. With the assumption of balance and symmetry, the ten phasors at a three phase port can be expressed in a much more simplified form as shown below.

L-N Voltages	L-L Voltages	Line Currents	Neutral Current
$\begin{cases} \vec{V}_{AN} = V_{LN} \angle \alpha_{AN} \\ \vec{V}_{BN} = V_{LN} \angle (\alpha_{AN} - 120^\circ) \\ \vec{V}_{CN} = V_{LN} \angle (\alpha_{AN} + 120^\circ) \end{cases}$	$\begin{cases} \vec{V}_{AB} = V_{LL} \angle \alpha_{AB} \\ \vec{V}_{BC} = V_{LL} \angle (\alpha_{AB} - 120^\circ) \\ \vec{V}_{CA} = V_{LL} \angle (\alpha_{AB} + 120^\circ) \end{cases}$	$\begin{cases} \vec{I}_A = I_L \angle \beta_A \\ \vec{I}_B = I_L \angle (\beta_A - 120^\circ) \\ \vec{I}_C = I_L \angle (\beta_A + 120^\circ) \end{cases}$	$\vec{I}_N = 0$

Notice that the neutral current is now shown to be zero. This is a special property seen at a three phase ports when a three phase system is balanced and symmetric. In addition to this, there are several more special

relationships. A list of the important mathematical relationships that exists at a three phase port due to balance and symmetry are listed below.

L-N voltage and L-L voltage magnitudes are related by the equation: $V_{LL} = \sqrt{3} \cdot V_{LN}$

L-L voltage phasors lead their associated L-N voltage phasors by 30°

Line current phasors lag their associated L-N voltage phasors by the power factor angle

The neutral current is zero

With the assumptions of balance and symmetry in three phase systems, a three phase port can now be completely described by only three simple quantities: line to line voltage magnitude, line current magnitude, and power factor angle. All port phasors can be determined from these three quantities, and also any three phase power calculation can be done given these three quantities.

Power at a three phase port is similar to power at a single phase port in that there are five different types of power that are defined (instantaneous, real, reactive, complex, and apparent). However the equations for these powers are obviously going to be different due to the basic difference between a single phase port and a three phase port. The table below defines the equations for three phase AC power quantities.

Quantity	Symbol	Equation(s)	Units
Instantaneous Power	$p_{3\phi}(t)$	$p_{3\phi}(t) = v_{AN}(t) \cdot i_A(t) + v_{BN}(t) \cdot i_B(t) + v_{CN}(t) \cdot i_C(t)$	Watts (W)
Real Power	$P_{3\phi}$	$P_{3\phi} = \sqrt{3} \cdot V_{LL} I_L \cos \theta$	Watts (W)
Reactive Power	$Q_{3\phi}$	$Q_{3\phi} = \sqrt{3} \cdot V_{LL} I_L \sin \theta$	Volt-Amperes Reactive (VAR)
Complex Power	$\bar{S}_{3\phi}$	$\bar{S}_{3\phi} = P_{3\phi} + jQ_{3\phi} = S_{3\phi} \angle \theta$ $P_{3\phi} = \text{Re}\{\bar{S}_{3\phi}\} \quad \text{and} \quad Q_{3\phi} = \text{Im}\{\bar{S}_{3\phi}\}$	Volt-Amperes (VA)
Apparent Power	$S_{3\phi}$	$S_{3\phi} = \bar{S}_{3\phi} = \sqrt{3} \cdot V_{LL} I_L = \sqrt{P_{3\phi}^2 + Q_{3\phi}^2}$	Volt-Amperes (VA)
Power Factor	pf	$pf = \cos \theta = \cos \left[\tan^{-1} (Q_{3\phi} / P_{3\phi}) \right]$	Unity, Leading, or Lagging ($\theta=0$), ($\theta<0$), ($\theta>0$)

Ohm's Law was used to yield additional power equations in single phase AC. However this is not done with three phase AC power for the obvious reason that a simple equivalent impedance (that has two terminals) does not exist at a three phase port (that has 3 or 4 terminals). Although there are methods for defining and working with special equivalent impedances at three phase ports, they will not be considered here.

One last important thing to mention about three phase electrical systems and three phase electrical power calculations are the standard assumptions made in three phase system specifications and final results. One should always remember the following two important rules.

1. Any voltage given in a three phase system or three phase electric machine specification is a line to line rms voltage magnitude quantity. Also any final result voltage answer must be a line to line rms voltage magnitude quantity.
2. Any power quantity given in a three phase system or three phase electric machine specification is a three phase power quantity. Also any final result power quantity answer must be a three phase power quantity.

Example #5:

A three phase Y-connected generator operates at 208V and supplies 20KVA at a 0.2 lagging power factor to a system. Calculate the generator current, and the real and reactive power output of the generator.

$$V_{LL} = 208 \text{ V}, \quad S_{3\phi} = 20 \text{ KVA}, \quad pf = 0.2 \text{ lagging} \rightarrow \theta = +\cos^{-1} 0.2 = 78.463^\circ$$

$$S_{3\phi} = \sqrt{3} \cdot V_{LL} I_L \rightarrow I_L = \frac{S_{3\phi}}{\sqrt{3} \cdot V_{LL}} = \frac{20,000}{\sqrt{3} \cdot 208} = 55.514 \text{ A} \leftarrow \text{generator output current}$$

$$P_{3\phi} = S_{3\phi} \cos \theta = 4000 \text{ W} = 4 \text{ KW} \leftarrow \text{generator real power output}$$

$$Q_{3\phi} = S_{3\phi} \sin \theta = 19,595.918 \text{ VAR} = 19.5969 \text{ KVAR} \leftarrow \text{generator reactive power output}$$

Example #6:

A three phase Δ -connected source outputs 20A, 10KW, and 50KVAR to a system. Calculate the source output voltage, output complex power, and output power factor.

$$I_L = 20 \text{ A}, \quad P_{3\phi} = 10 \text{ KW}, \quad Q_{3\phi} = 50 \text{ KVAR}$$

$$\bar{S}_{3\phi} = S_{3\phi} \angle \theta = P_{3\phi} + jQ_{3\phi} = 50,990.1951 \angle 78.690^\circ \text{ VA} \leftarrow \text{source complex power output}$$

$$pf = \cos \theta = 0.196116 \text{ lagging } (\theta > 0) \leftarrow \text{source output power factor}$$

$$S_{3\phi} = \sqrt{3} \cdot V_{LL} I_L \rightarrow V_{LL} = \frac{S_{3\phi}}{\sqrt{3} \cdot I_L} = \frac{50,990.1951}{\sqrt{3} \cdot 20} = 1471.960 \text{ V} \leftarrow \text{source output voltage}$$

Example #7:

A three phase Y-connected load operates at 4162V, 180KW, and a 0.75594 lagging power factor. Calculate the load input current and complex power input.

$$V_{LL} = 4162 \text{ V}, \quad P_{3\phi} = 180 \text{ KW}, \quad pf = 0.75594 \text{ lagging} \rightarrow \theta = +\cos^{-1} 0.75594 = 40.892^\circ$$

$$P_{3\phi} = \sqrt{3} \cdot V_{LL} I_L \cos \theta \rightarrow I_L = \frac{P_{3\phi}}{\sqrt{3} \cdot V_{LL} \cos \theta} = \frac{180,000}{\sqrt{3} \cdot 4162 \cdot \cos 40.892^\circ} = 33.031 \text{ A} \leftarrow \text{load input current}$$

$$P_{3\phi} = S_{3\phi} \cos \theta \rightarrow S_{3\phi} = \frac{P_{3\phi}}{\cos \theta} = \frac{P_{3\phi}}{pf} = \frac{180,000}{0.75594} = 238,114.1360 \text{ VA}$$

$$\bar{S}_{3\phi} = S_{3\phi} \angle \theta = 238.1141 \angle 40.892^\circ \text{ KVA} \leftarrow \text{load complex power input}$$

Additional Practice Problems

Below are given several practice problems requiring basic single phase and three phase interface and power calculations. The answers are given in parenthesis with each problem so that you can check your work.

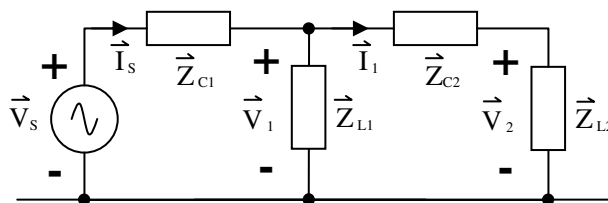
1. A single phase 240V source supplies 5KW and 500VAR to a constant impedance load. Compute the following quantities.
 - a. Complex power supplied by the source. (5.0249 $\angle$ 5.711 $^\circ$ KVA)
 - b. Source power factor. (0.995037 lagging)
 - c. Source current. (20.937A)
 - d. Equivalent impedance of the load. (11.4628 $\angle$ 5.711 $^\circ\Omega$)
2. A single phase 120V source supplies 40A with a 0.985 leading power factor to a constant impedance load. Compute the following quantities.
 - a. Complex power supplied by the source. (4.8 $\angle$ -9.936 $^\circ$ KVA)
 - b. Real and reactive power supplied by the source. (4.728KW and -828.261VAR)
 - c. Equivalent impedance of the load. (3 $\angle$ -9.936 $^\circ\Omega$)
3. A single phase 960V source supplies 180KVA with a 0.8 lagging power factor to a constant impedance load. Compute the following quantities.

- a. Complex power supplied by the source. ($180\angle 36.870^\circ \text{KVA}$)
- b. Real and reactive power supplied by the source. (144KW and 108KVAR)
- c. Source current. (187.5A)
- d. Equivalent impedance of the load. ($5.12\angle 36.870^\circ \Omega$)

4. A single phase 480V source supplies 75KW with a 0.975 leading power factor to a constant impedance load. Compute the following quantities.
 - a. Source current. (160.256A)
 - b. Complex power supplied by the source. ($76.9231\angle -12.839^\circ \text{KVA}$)
 - c. Reactive power supplied by the source. (-17.0927KVAR)
 - d. Equivalent impedance of the load. ($2.9952\angle -12.839^\circ \Omega$)

5. A single phase 208V source supplies 4KVAR with a 0.775 lagging power factor to a constant impedance load. Compute the following quantities.
 - a. Source current. (30.430A)
 - b. Complex power supplied by the source. ($6.3295\angle 39.195^\circ \text{KVA}$)
 - c. Real power supplied by the source. (4.9054KW)
 - d. Equivalent impedance of the load. ($6.8353\angle 39.195^\circ \Omega$)

6. The AC circuit shown below models a small power system. The circuit parameters are given and also shown below. Calculate the following quantities in the circuit.

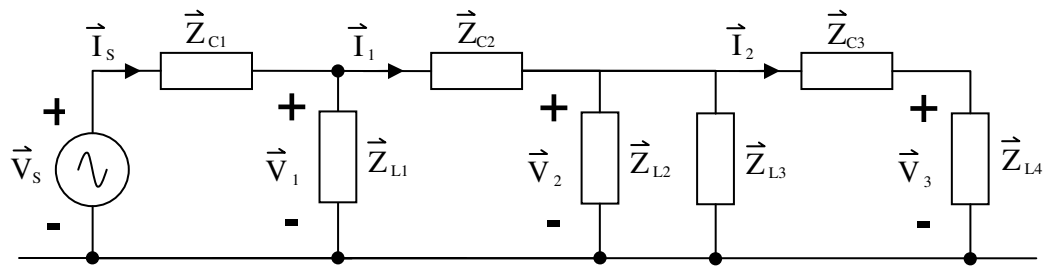


$$\begin{aligned}\vec{V}_s &= 125\angle 0^\circ \text{ V} \\ \vec{Z}_{L1} &= 300\angle -5^\circ \Omega, \quad \vec{Z}_{L2} = 20\angle 75^\circ \Omega \\ \vec{Z}_{C1} &= 1\angle 25^\circ \Omega, \quad \vec{Z}_{C2} = 1.25\angle 25^\circ \Omega\end{aligned}$$

- a. Phasor voltages $\mathbf{V}_1$ and $\mathbf{V}_2$. ($120.661\angle 1.862^\circ \text{V}$, $115.878\angle 4.497^\circ \text{V}$)
- b. Phasor currents $\mathbf{I}_s$ and $\mathbf{I}_1$. ($5.895\angle -66.685^\circ \text{A}$, $5.794\angle -70.503^\circ \text{A}$)
- c. Apparent power supplied by the source. (736.872VA)
- d. Energy in KWH supplied by the source in a 30-day period. (209.980KWH)
- e. Real power consumed by impedance $\mathbf{Z}_{C1}$. (31.495W)
- f. Reactive power consumed by impedance $\mathbf{Z}_{L1}$. (-4.230VAR)
- g. Complex power consumed by impedance $\mathbf{Z}_{L2}$. ($671.390\angle 75^\circ \text{VA}$)
- h. Energy in KWH consumed by impedance $\mathbf{Z}_{L2}$ in a 20-week period. (583.862KWH)

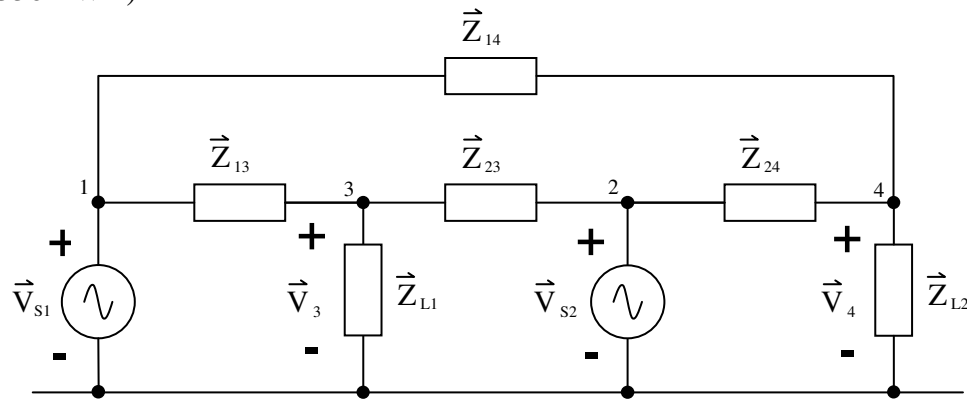
7. The AC circuit shown below models a small power system. The circuit parameters are given and also shown below. Calculate the following quantities in the circuit.

- a. Phasor voltages $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$. ($200.843\angle -5.156^\circ \text{V}$, $188.807\angle -2.561^\circ \text{V}$, $186.369\angle -1.943^\circ \text{V}$)
- b. Phasor currents $\mathbf{I}_s$, $\mathbf{I}_1$, and $\mathbf{I}_2$. ($43.856\angle -0.699^\circ \text{A}$, $11.935\angle -65.102^\circ \text{A}$, $3.727\angle -66.943^\circ \text{A}$)
- c. Real and reactive power supplied by the source. (10.5246KW , 0.1284KVAR)
- d. Energy in KWH supplied by the source in a 3-day period. (757.772KWH)
- e. Real power consumed by impedance $\mathbf{Z}_{C1}$. (1743.130W)
- f. Real power consumed by impedance $\mathbf{Z}_{C3}$. (10.703W)
- g. Reactive power consumed by impedance $\mathbf{Z}_{L1}$. (-2.7593KVAR)
- h. Complex power consumed by impedance $\mathbf{Z}_{L2}$. ($891.206\angle 75^\circ \text{VA}$)
- i. Complex power consumed by impedance $\mathbf{Z}_{L4}$. ($694.671\angle 65^\circ \text{VA}$)
- j. Energy in KWH consumed by impedance $\mathbf{Z}_{L3}$ in a 9-week period. (762.263KWH)



$$\begin{aligned}\vec{V}_s &= 240\angle 0^\circ \text{ V}, \quad \vec{Z}_{C1} = 1\angle 25^\circ \Omega, \quad \vec{Z}_{C2} = 1.25\angle 25^\circ \Omega, \quad \vec{Z}_{C3} = 0.85\angle 25^\circ \Omega \\ \vec{Z}_{L1} &= 5\angle -20^\circ \Omega, \quad \vec{Z}_{L2} = 40\angle 75^\circ \Omega, \quad \vec{Z}_{L3} = 50\angle 45^\circ \Omega, \quad \vec{Z}_{L4} = 50\angle 65^\circ \Omega\end{aligned}$$

8. The AC circuit shown below models a small power system. The circuit parameters are given and also shown below. Calculate the following quantities in the circuit.
- Phasor voltages $\vec{V}_3$ and $\vec{V}_4$. (79.3818 $\angle$ -0.566°KV, 79.1395 $\angle$ -0.696°KV)
 - Real and reactive power supplied by the source connected to node 1. (17.279013MW, 9.040393MVAR)
 - Real and reactive power supplied by the source connected to node 2. (8.516595MW, 23.675259MVAR)
 - Real and reactive power consumed by impedance $\vec{Z}_{L1}$. (7.876834MW, 13.643078MVAR)
 - Real and reactive power consumed by impedance $\vec{Z}_{L2}$. (17.71462MW, 17.71462MVAR)
 - Total combined energy in MWH consumed by impedances $\vec{Z}_{L1}$ and $\vec{Z}_{L2}$ in a 5-day period. (3070.974556MWH)



$$\begin{aligned}\vec{V}_{S1} &= 80,540\angle 1.5^\circ \text{ V}, \quad \vec{V}_{S2} = 81,406\angle 0^\circ \text{ V}, \quad \vec{Z}_{L1} = 400\angle 60^\circ \Omega, \quad \vec{Z}_{L2} = 250\angle 45^\circ \Omega \\ \vec{Z}_{13} &= 11.803\angle 82.91^\circ \Omega, \quad \vec{Z}_{23} = 23.607\angle 82.91^\circ \Omega, \quad \vec{Z}_{14} = 34.538\angle 80.72^\circ \Omega, \quad \vec{Z}_{24} = 10.361\angle 80.72^\circ \Omega\end{aligned}$$

9. A three phase 480V source supplies 25KW and 9800VAR to a power system. Using source voltage $\vec{V}_{BC}$ as a reference phasor, compute the following quantities.
- Complex power supplied by the source. (26.8522 $\angle$ 21.405°VA)
 - Source power factor. (0.931023 lagging)
 - Source line to line voltage phasors. (480 $\angle$ 120°V, 480 $\angle$ 0°V, 480 $\angle$ -120°V)
 - Source line to neutral voltage phasors. (277.128 $\angle$ 90°V, 277.128 $\angle$ -30°V, 277.128 $\angle$ -150°V)
 - Source line current phasors. (32.298 $\angle$ 68.595°A, 32.298 $\angle$ -51.405°A, 32.298 $\angle$ -171.405°A)
10. A three phase 13.8KV source supplies 1.95MVA with a 0.985 leading power factor to a power system. Using source current $\vec{I}_A$ as a reference phasor, compute the following quantities.
- Complex power supplied by the source. (1.95 $\angle$ -9.936°MVA)
 - Real and reactive power supplied by the source. (1.92075MW, -0.336481MVAR)
 - Source line to line voltage phasors. (13.8 $\angle$ 20.064°KV, 13.8 $\angle$ -99.936°KV, 13.8 $\angle$ 140.064°KV)
 - Source line to neutral voltage phasors. (7.967 $\angle$ -9.936°KV, 7.967 $\angle$ -129.936°KV, 7.967 $\angle$ 110.064°KV)
 - Source line current phasors. (81.582 $\angle$ 0°A, 81.582 $\angle$ -120°A, 81.582 $\angle$ 120°A)

11. A three phase 4160V Δ -connected source supplies 50A with a 0.78 lagging power factor to a Y-connected constant impedance load. Using source voltage $\vec{V}_{CN}$ as a reference phasor, compute the following quantities.
- Complex power supplied by the source. (360.2666 $\angle$ 38.739°KVA)
 - Real and reactive power supplied by the source. (281.0079KW, 225.4474KVAR)
 - Source line to line voltage phasors. (4160 $\angle$ -90°V, 4160 $\angle$ 150°V, 4160 $\angle$ 30°V)
 - Source line to neutral voltage phasors. (2401.777 $\angle$ -120°V, 2401.777 $\angle$ 120°V, 2401.777 $\angle$ 0°V)
 - Source line current phasors. (50 $\angle$ -158.739°A, 50 $\angle$ 81.261°A, 50 $\angle$ -38.739°A)
 - Source Δ -phase current phasors. (28.868 $\angle$ -128.739°A, 28.868 $\angle$ 111.261°A, 28.868 $\angle$ -8.739°A)
 - Per phase equivalent impedance of the Y-connected load. (48.0355 $\angle$ 38.739° Ω)
-
12. A three phase 12KV Y-connected source supplies power to three different loads connected in parallel. The source supplies 875KW and 205KVAR. The first load is Y-connected and absorbs 99KVA at a 0.8258 lagging power factor. The second load is Y-connected and absorbs 520KVA at a 0.5195 leading power factor. The third load is Δ -connected, and its power consumption is unknown. Compute the following quantities.
- Real and reactive power absorbed by the third load. (523.1058KW, 593.4923KVAR)
 - Per phase equivalent impedances of the loads. (1454.5455 $\angle$ 34.330° Ω , 276.9231 $\angle$ -58.701° Ω , 546.0604 $\angle$ 48.607° Ω)
 - Source line current phasors using source voltage $\vec{V}_{AN}$ as a reference phasor. (43.238 $\angle$ -13.186°A, 43.238 $\angle$ -133.186°A, 43.238 $\angle$ 106.814°A)
 - First load line current phasors using source voltage $\vec{V}_{AN}$ as a reference phasor. (4.763 $\angle$ -34.330°A, 4.763 $\angle$ -154.330°A, 4.763 $\angle$ 85.670°A)
 - Second load line current phasors using source voltage $\vec{V}_{AN}$ as a reference phasor. (25.019 $\angle$ 58.701°A, 25.019 $\angle$ -61.299°A, 25.019 $\angle$ 178.701°A)
 - Third load line current phasors using source voltage $\vec{V}_{AN}$ as a reference phasor. (38.063 $\angle$ -48.607°A, 38.063 $\angle$ -168.607°A, 38.063 $\angle$ 71.393°A)
 - Third load Δ -phase current phasors using source voltage $\vec{V}_{AN}$ as a reference phasor. (21.976 $\angle$ -18.607°A, 21.976 $\angle$ -138.607°A, 21.976 $\angle$ 101.393°A)